Journal of Molecular Structure 1069 (2014) 122–126

Contents lists available at ScienceDirect

Journal of Molecular Structure

journal homepage: www.elsevier.com/locate/molstruc

Two-dimensional correlation analysis to study variation of near-infrared water absorption bands in the presence of inorganic acids

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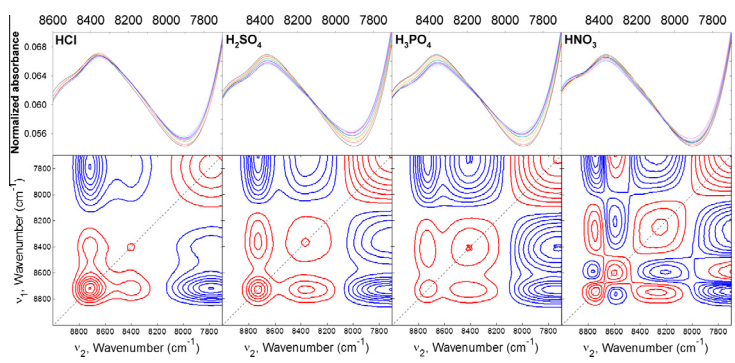
HIGHLIGHTS

- NIR water band was varied by presence HCl, H_2SO_4, H_3PO_4, HNO_3 was investigated by 2D-COS.
- The variation was due to interaction of dissociated ions with water vibration.
- 2D COS revealed that individual acids differently influenced water vibration.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:
Available online 8 December 2013

Keywords:
Etchant solution
Inorganic acid
Varied water band
NIR spectroscopy
2D correlation spectroscopy

ABSTRACT

Two-dimensional (2D) correlation analysis has been utilized to investigate NIR water bands perturbed by the presence of four different inorganic acids individually: HCl, H_2SO_4, H_3PO_4, and HNO_3. The observed spectral variation in the 9000–7700 cm^{-1} range was mainly due to interaction of dissociated H_3O^+ and corresponding anions with the vibration of water in a hydrogen bonding network. 2D correlation analysis of NIR spectra acquired from sample solutions (concentration range: 0.2–1.0 M) showed that individual acids differently influenced water vibration. In addition, unforeseen spectral variations under the water band that were difficult to identify with corresponding raw NIR spectra were clearly observed. Based on the asynchronous correlation analysis, three underlying individual variations occurred for HCl under the 8718 cm^{-1} band. Only two asynchronous correlations were observed for H_2SO_4 and H_3PO_4. The 2D correlation features of HNO_3 were distinctly different from those of the other three acids due to an additional spectral feature caused by direct absorption by NO_3^-. The dissimilar influence of the selected acids on water vibration was confirmed by NIR spectroscopy combined with 2D correlation analysis. Partial least squares (PLS) loadings from each case were compared to examine the difference in weights that were constructed to follow the corresponding concentration changes.

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1. Introduction

Near-infrared (NIR) spectroscopy has been demonstrated as a viable analytical method to measure concentrations of components in diverse wafer etchant solutions [1,2] composed of an oxidizing agent (H_2O_2) and acids. Usually, strong inorganic acids such as HCl completely dissociate into hydronium ions (H_3O^+) and a corresponding anion in an aqueous solution. The dissociated species directly absorb NIR radiation and/or interact with water hydrogen boding networks to alter the NIR spectral features of water. In our previous publications, HCl and HF concentrations were measured in etchant solutions by recognizing water bands [1–3] rather than direct NIR absorption [4–6]. The altered water spectral features

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http://dx.doi.org/10.1016/j.molstruc.2013.12.005

were found to correlate with corresponding concentration changes using multivariate calibration methods such as partial least squares (PLS).

Because water spectral features vary without a distinct signature, the component-specificity and robustness of a PLS model are of interest. An in-depth investigation on the effects of dissolved inorganic acids on the NIR spectral features of water has not been performed even though perturbed water bands are a valuable source for determining concentration. More profound investigations of water band variation are necessary to validate the selectivity (component-specificity) of a PLS model.

This publication investigated the variation in NIR water bands with dissolved inorganic acids using generalized two-dimensional (2D) correction analysis. For this purpose, four different inorganic acids (HCl, H_2SO_4, H_3PO_4, and HNO_3) were selected, and the NIR spectra of acid solutions were collected within a concentration range of 0.2–1.0 M. 2D correlation analysis was then performed to investigate the water bands perturbed by each acid, and the dissimilarity in perturbations was examined. A cross-validated PLS model was built using each spectral dataset, and corresponding PLS loadings were compared to identify the difference in weights utilized to describe concentration changes.

2. Experimental

High purity HCl, H_2SO_4, H_3PO_4, and HNO_3 were purchased from a common supplier. For each acid, nine solutions with concentrations ranging from 0.2 to 1.0 M (increment intervals of 0.1 M) were prepared, and NIR spectra (resolution: 8 cm^{-1}) were collected using an ABB Bomem MB-160 FT-NIR spectrometer (Quebec City, Quebec, Canada) equipped with a tungsten–halogen source and InGaAs detector. The samples were transferred into a glass vial inserted into a heated metal vial holder that could control temperature with a precision of ±0.1 °C. The temperature of all samples was maintained at 25.0 °C during spectral collection. Air was used as background for absorbance spectra. All spectral pre-processing and multivariate analyses, including baseline correction, normalization, 2D correlation analysis, and PLS, were performed using MATLAB Version 7.0 (Math Works Inc., MA, USA).

3. Results and discussion

3.1. NIR spectral features of HCl, H_2SO_4, H_3PO_4, and HNO_3 solutions

Fig. 1(a–d) shows normalized NIR spectra (9000–7700 cm^{-1} range) of HCl, H_2SO_4, H_3PO_4, and HNO_3, respectively, within a concentration range of 0.2–1.0 M. For normalization, the baselines of raw spectra were vertically corrected and zeroed at 9000 cm^{-1}. The baseline-corrected spectra were then divided by the corresponding integrated areas calculated in the 9000–7700 cm^{-1} range [7,8]. For comparison, the NIR spectrum of pure water, corresponding to the second overtone of the –OH vibration, was included in each plot. For better analysis, spectral variation in the 8600–7700 cm^{-1} range was also highlighted in each figure. The overall spectral features are similar to those of water for each case, and minute spectral variation occurred around the water band according to the change in acid concentration. This indicates that the four acids do not have distinct absorption features. To examine spectral variation in detail, difference spectra were obtained by subtracting individual acid spectra from a pure water spectrum. Difference spectra are displayed for HCl, H_2SO_4, H_3PO_4, and HNO_3 in Fig. 1(e–h), respectively.

HCl completely dissociates into H_3O^+ and Cl^-. Since Cl^- does not absorb NIR radiation, the observed spectral variation originated from the presence of generated H_3O^+ and Cl^-. These ions caused displacement of water molecules and/or perturbation of the water band, resulting in variation of water spectral features. As shown, two different types of spectral variation are noticeable with changes in HCl concentration. The magnitude of water absorption decreased with increased HCl concentration in the 9000–8300 cm^{-1} range, while it increased simultaneously in the 8200–7700 cm^{-1} range. The positive and negative scales in the difference spectra indicate decreased and increased water absorption, respectively. The decrease in absorption in the 9000–8300 cm^{-1} range is due to displacement of water molecules caused by H_3O^+ and Cl^-. As previously studied, water displacement always occurs when foreign components are added into pure water [8–10]. The increase of water absorption in the 8200–7700 cm^{-1} range originated from broadening of the water band. Absorption features in this range correspond to the left edge of the highly absorbing second overtone water band centered at 6800 cm^{-1}. When charged species are present in water, they interact with the water hydrogen bonding network and increase diversity in water vibration. This results in band broadening, as indicated in previous publications [8,10]. Therefore, increased absorption is due to the widened bandwidth of the second overtone water band caused by greater interaction with dissociated species.

Spectral variation with H_2SO_4 is generally similar to that with HCl, although the spectral difference around 8400 cm^{-1} is more significant and the magnitude of spectral variation in the range of 8200–7700 cm^{-1} is greater for H_2SO_4. Spectral variations around 8700 and 8400 cm^{-1} become less and more significant, respectively, for H_3PO_4. The variation occurring around 8400 cm^{-1} is likely due to perturbed water vibration caused by large ions such as HSO_4^- and $H_2PO_4^-$, which are the predominant anion species in H_2SO_4 and H_3PO_4 solutions, respectively. In contrast, the spectral variation around 8400 cm^{-1} is minimal for HCl. The second acid dissociation constant (Ka_2) of H_2SO_4 is 1.03×10^{-2}, indicating a considerable amount of SO_4^{2-} (10.7–25.4% relative to that of HSO_4^-) in the same matrix, and SO_4^{2-} also participates in the interaction with water molecules. The second (Ka_2) and third acid dissociation constants (Ka_3) of H_3PO_4 are 6.34×10^{-8} and 4.22×10^{-13}, respectively, indicating that the concentrations of HPO_4^{2-} ($7.48 \times 10^{-5} \sim 0.135\%$ relative to that of $H_2PO_4^-$) and PO_4^{3-} are very low, and their influences on water spectral features are insignificant. Due to differences in the structure, size, and charge of dissociated anion species from H_2SO_4 and H_3PO_4, their perturbations of water vibration are dissimilar, resulting in different spectral variations.

When two mono-protic acids (HCl and HNO_3) are compared, their spectral variations are clearly different. The spectral difference around 8300 cm^{-1} for HNO_3 is due to the presence of NO_3^-, a larger anion than Cl^-. Another noticeable feature in the HNO_3 spectra is increased absorption at 8600 cm^{-1} due to the direct absorption of NO_3^- (second overtone of NO stretching) [7,11]. The dissociated anions from H_2SO_4 and H_3PO_4 do not absorb NIR radiation in this range. Overall, NIR spectral features differ with varying HCl, H_2SO_4, H_3PO_4, and HNO_3 concentrations.

3.2. Two-dimensional (2D) correlation analysis to examine water band variation with changes in HCl, HNO_3, H_2SO_4, and H_3PO_4 concentrations

Generalized 2D correlation analysis is helpful for identifying unforeseen and minute spectral variation that is difficult to recognize in original raw spectra [12]. 2D correlation analysis can effectively capture spectral feature changes in response to an applied perturbation, such as increases/decreases in intensity and shifted band position. Nine NIR spectra in the concentration range from 0.2 to 1.0 M for each acid were used for 2D correlation analysis. The applied perturbation was increased interaction with water

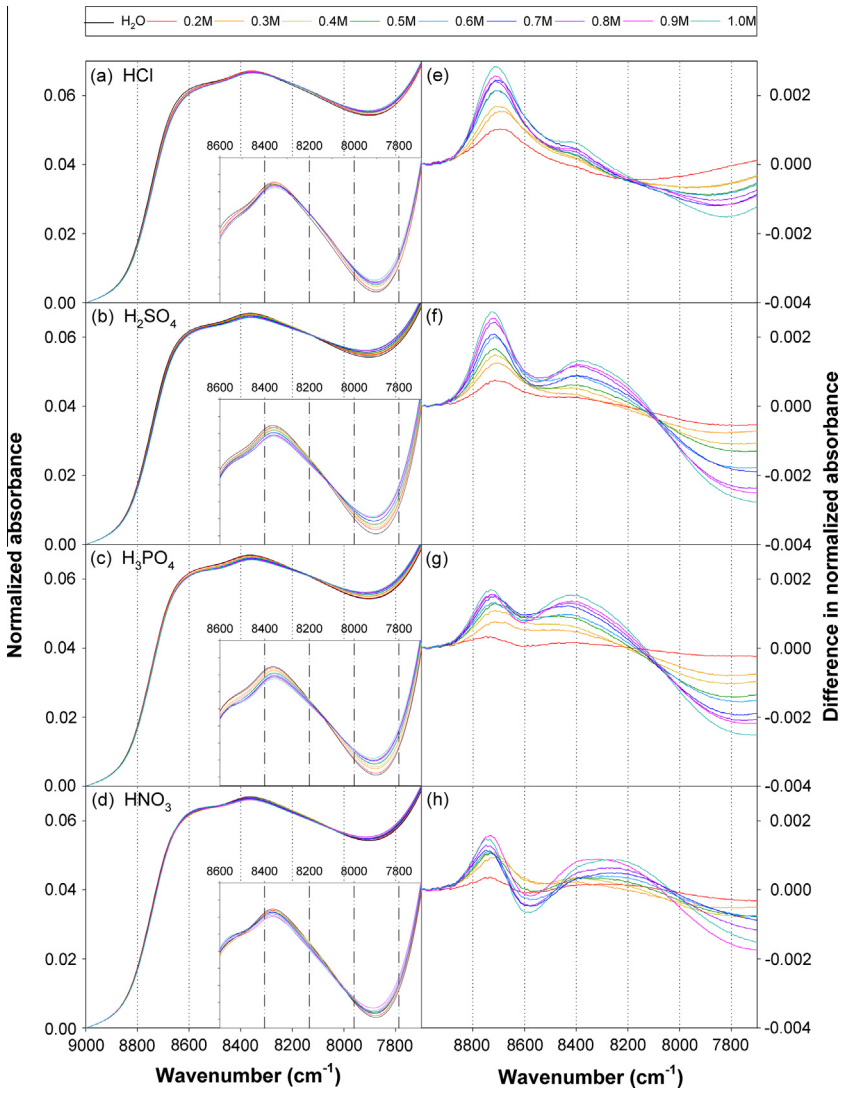

Fig. 1. Normalized NIR spectra (9000–7500 cm^{-1} range) of HCl (a), H_2SO_4 (b), H_3PO_4 (c), and HNO_3 (d) in the concentration range of 0.2–1.0 M. Difference spectra obtained by subtracting individual acid spectrum from pure water spectrum for HCl (e), H_2SO_4 (f), H_3PO_4 (g), and HNO_3 (h).

vibration caused by individual acids, more specifically by their dissociated ionic species.

Fig. 2 shows synchronous (top row) and asynchronous (bottom row) 2D correlation spectra obtained using each NIR spectral dataset of HCl, H_2SO_4, H_3PO_4, and HNO_3. Red and blue colors indicate positive and negative correlations observed with increases in concentration, respectively. Most notably, the synchronous 2D correlation spectrum of HNO_3 distinctly differs from those of the three other acids due to the NO_3^- absorption band. The 2D correlation features of HCl, H_2SO_4, and H_3PO_4 are dissimilar, indicating that each dissociated ion interacts uniquely with water molecules in a hydrogen bonding network.

For HCl, the 8718 and 7788 cm^{-1} bands vary significantly with increased concentration. The 8718 cm^{-1} band is composed of three underlying individual variations at 8870, 8740, and 8615 cm^{-1}, as revealed in the corresponding asynchronous correlation spectrum. In the asynchronous correlation spectra of H_2SO_4 and H_3PO_4, only two asynchronous correlation peaks are observable under the 8725 cm^{-1} band. This indicates that the perturbation of HCl on the water vibration band at 8718 cm^{-1} is not analogous to those of H_2SO_4 and H_3PO_4. This result is difficult to observe by examination of the corresponding raw or difference spectra. With varied H_2SO_4 concentrations, the correlation is less and more dominant at 8725 and 8363 cm^{-1}, respectively, compared to the HCl spectrum. This trend is even more apparent with H_3PO_4. The 2D correlation analysis results suggest that anions dissociated from HCl, H_2SO_4, and H_3PO_4 differently influence water vibration. These dissimilar interactions allow NIR spectroscopic differentiation of the three acids, which show no NIR absorption themselves.

3.3. Examination of PLS loading built to determine acid concentration

Along with 2D correlation analysis, the examination of loading in a PLS model is informative for identifying significant spectral features related to concentration variations in each acid. For this purpose, PLS models were developed for each case by fully cross-validating all samples in a dataset. Of nine total samples, six and three samples were randomly assigned into calibration and validation sets, allowing 84 combinations to select six samples for calibration ($_9C_6 = 84$ combinations). For each combination, a standard error of calibration (SEC) and standard error of prediction (SEP) were calculated. Finally, the mean standard error of calibration (MSEC) and mean standard error of prediction (MSEP) were calculated by averaging the SEC and SEP values obtained from all combinations, respectively. In all cases, two PLS factors were used. The resulting MSEPs for the determination of HCl, H_2SO_4, H_3PO_4,

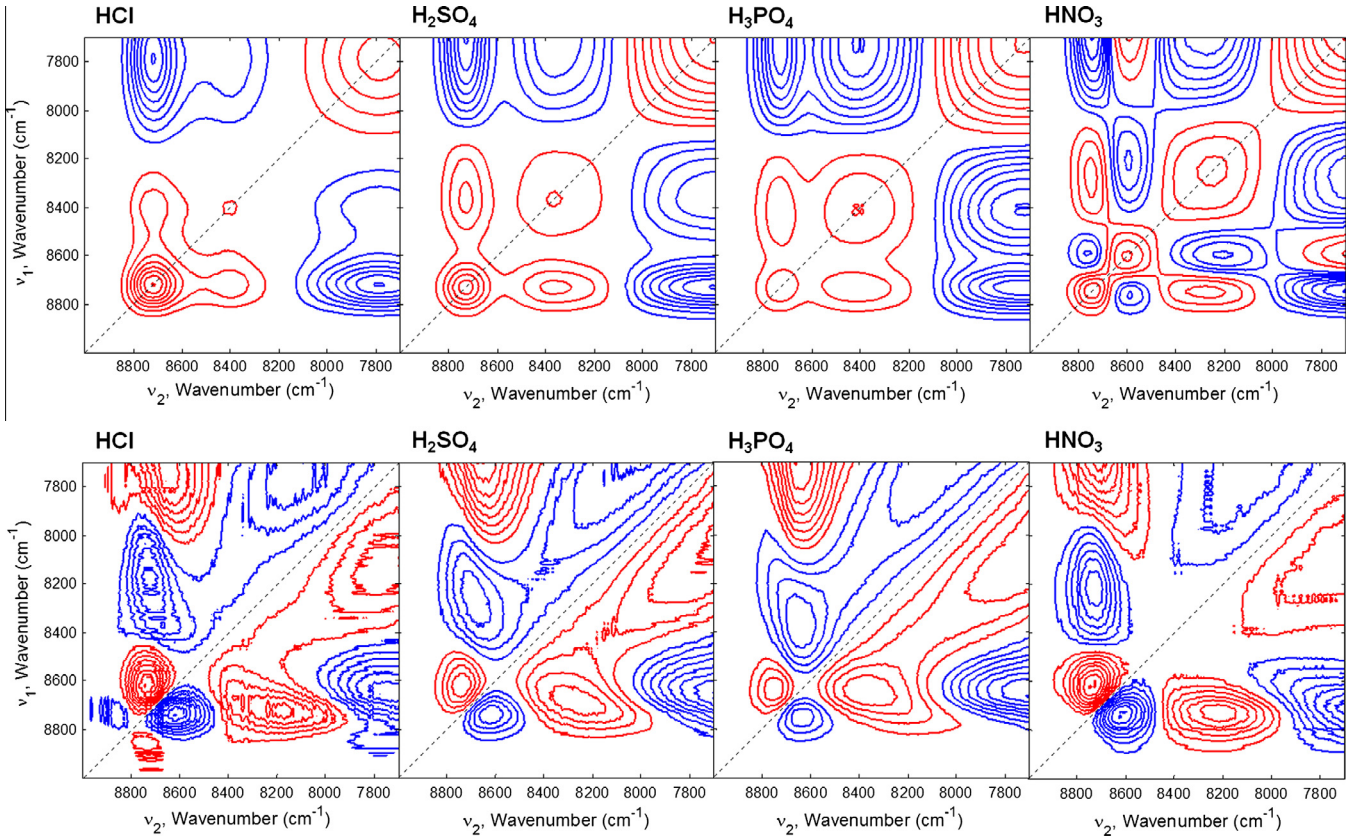

Fig. 2. Synchronous (top row) and asynchronous (bottom row) 2D correlation spectra obtained using each NIR spectral dataset of HCl, H_2SO_4, H_3PO_4, and HNO_3. Red and blue indicate positive and negative correlations observed with increased acid concentration, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and HNO_3 concentrations were 0.06, 0.03, 0.03, and 0.06 M, respectively. There is no significant difference among MSEPs from each case, while MSEPs for determining H_2SO_4 and H_3PO_4 concentrations were slightly lower than those of HCl and HNO_3. The relatively larger degree of spectral variation (or perturbation on water vibration) induced by these polyprotic acids is responsible for the better accuracy.

Fig. 3(a) and (b) shows the first and second loadings, respectively, used in the corresponding PLS models to determine each acid concentration. The first loading of HNO_3 is substantially different from those of the three other acids, again confirming that the NO_3^- absorption around 8600 cm^{-1} correlates with concentration variation in HNO_3. In contrast, the first loading features of HCl, H_2SO_4, and H_3PO_4 are closer to each other, such as the similar

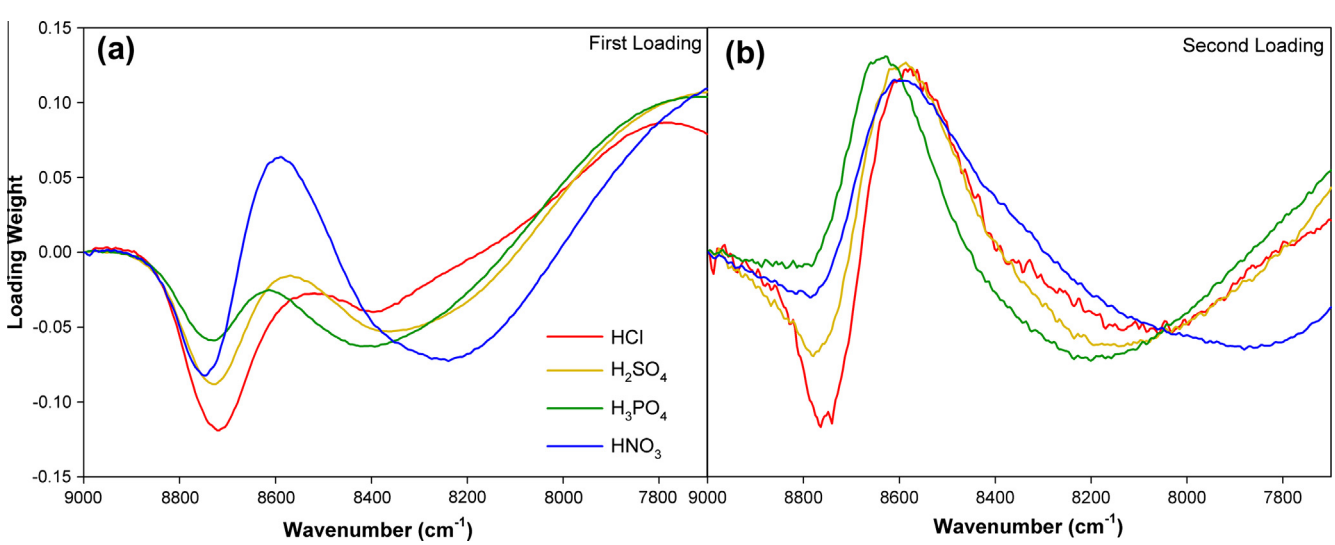

Fig. 3. First (a) and second loadings (b) used in PLS models to determine HCl (red), H_2SO_4 (yellow), H_3PO_4 (green), and HNO_3 (blue) concentrations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

loading weight on 8720 cm^{-1} in all cases. For the 8700–8400 cm^{-1} range, weight positions are different from each other. When the second loadings of HCl, H_2SO_4, and H_3PO_4 are compared for mono- to tri-protic acid, the weight on 8780 cm^{-1} is decreased and the weight on 8600 cm^{-1} shifts to a slightly higher wavenumber. The second loading of HNO_3 mainly weighs the 8600 cm^{-1} band for modeling, similar to the first loading. Even for species (HCl, H_2SO_4, and H_3PO_4) unable to absorb NIR radiation, loading features are different from each other, clearly indicating their dissimilar physical behaviors in water.

4. Conclusions

2D correlation analysis was used to study variation in NIR water band caused by the presence of HCl, H_2SO_4, H_3PO_4, and HNO_3, which are all major components in etchant solutions. Although HCl, H_2SO_4, and H_3PO_4 did not directly absorb NIR radiation, water spectral features varied with these acids, indicating a valuable source for determining their concentrations. Because some etchant solutions contain more than two acids, it is valuable to examine their perturbation on water spectral features and understand their interactions with water. Since more than two acids are simultaneously present, resulting NIR features are less characteristic of individual components. PLS calibration for a given acid and corresponding accuracy with increased compositional complexity (number of acid components) are of interest in such systems. The above-mentioned issues are currently under investigation.

Acknowledgment

This research was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology (2012R1A1B3003965).

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